MFE 230GB · CURRENCY MARKETS · FINAL PROJECT · 8 OCTOBER 2026

Equity Spillover into FX

When a country's stock market beats the US, investors buy its assets and its currency follows. We trade that flow in G10 and in emerging markets.

Rationale

FX demand is inelastic and mostly latent (Topic 5). Equity performance pulls foreign flow, and the dollar trades with US equities (Topic 3). The flow is absorbed over weeks.

Strategy

Each month, each currency: long against USD if its stock index beat the US over six months, short if it lagged. Equal weight. G10 at 2 bp, nine floating EM currencies at 15 bp.

Performance

G10: Sharpe 0.41 since 2000 after costs, 0.46 in the selection window and 0.38 after, max drawdown 9%. EM floats: 0.36, and 0.24 after 2016, but mostly a dollar bet.

1 · THE IDEA

A stock market that beats the US attracts flow, and flow is currency demand

Demand sets the exchange rate

Latent demand is 82% of FX variation and elasticities are 1 to 30 (Topic 5). A foreign investor who wants a country's stocks buys its currency first, and financial flow pushes price (Topic 4).

The dollar is the other side

The dollar is on 89% of trades (Topic 1) and since 2025 tracks US equity outperformance more than rate gaps (Topic 3). So the question is relative: did the market beat the US, not did it go up.

The sign is an empirical question

Hau and Rey's uncovered equity parity predicts the opposite: investors rebalance out of winners. If the flow story is right, the effect is positive and short-lived, absorbed within weeks (Topic 4).

THREE FACTS, 2000 TO 2026

0.15 / 0.30
same-month correlation of a country's equity return and its currency, G10 / EM

0.41 vs −0.03
Sharpe when the signal is "beat the US" versus "went up": positive, and relative

0.12
Sharpe if the signal is used one month late: the flow is absorbed within weeks

Monthly; OECD share-price indices in local currency; FRED, ECB and IMF exchange rates; OECD 3-month rates for carry where available · 2

2 · How we built it: two families of construction

Family 1 · Cross-sectional: rank countries against each other

Score every currency each month, long the strongest group, short the weakest. Longs equal shorts, so the book is dollar-neutral. Four choices, 64 combinations:

Choice	Options
Measure	**Local:** the index in its own currency. **USD:** converted to dollars, so the currency move is in the signal. **Over cash:** local minus the 3-month rate. **Vol-adjusted:** local divided by trailing volatility
Lookback	1, 3, 6 or 12 months
Weighting	**Tercile:** equal weight in the top and bottom third. **Rank:** weight proportional to rank, the middle held in small size
Holding	Monthly, or three months with overlapping books

Family 2 · Dollar-relative

No ranking. Each currency is judged against the US alone: beat the US over the lookback, long it against the dollar; lagged, short it. Equal weight.

Not dollar-neutral: if six of nine markets beat the US the book is net long foreign currency; if the US beats everyone, it is long the dollar against all nine.

Lookback 3, 6 or 12 months. **Measure** relative to the US, or own return. Six combinations.

The difference: Family 1 compares countries with each other and holds equal longs and shorts, so no dollar position. Family 2 compares each country with the US only, so the longs and shorts need not balance and the book carries a net dollar position.

Same for every version: month-end rebalance, local-currency OECD indices, excess returns with carry, 2 bp half-spread in G10 and 15 bp in EM. Seventy constructions in all · 3

3 · Selection: chosen on 2000 to 2015, judged on 2016 to 2026

FOUR CONSTRUCTIONS, G10, 6-MONTH LOOKBACK, SHARPE AFTER 2 BP COSTS

Construction	How the book is built each month	2000–15	2016–26	Full
A · Cross-section, top 3 / bottom 3	Rank the nine on 6-month local return. Long the top three, short the bottom three, equal weight. Longs equal shorts	0.55	0.06	0.37
B · Cross-section, ranked	Same ranking on vol-adjusted return. All nine held, weight proportional to rank, best long, worst short. Longs equal shorts	0.70	0.07	0.46
C · Dollar-relative vs US: the book	No ranking. Each currency: 6-month local return minus the US. Positive, long vs USD; negative, short. Weight 1/9 each	**0.46**	**0.38**	**0.41**
D · Dollar-relative, own return	Same as C but the signal is the market's own 6-month return, with no US benchmark. Up, long; down, short	0.09	−0.24	−0.03

Cross-section vs dollar-relative

Cross-section (A, B) compares countries with each other: a market that lagged the US can still be a long if the others lagged more. Longs equal shorts, so no dollar position. **Dollar-relative (C, D)** compares each country with the US only: long if it beat the US, short if not, however many fall on each side. The book can be net long or short dollars.

The rule we set before looking

Choose on 2000 to 2015 only; report 2016 to 2026 untouched. A and B look good to 2015, then go flat; B was the best of all 64 in sample. C holds in both halves. D drops the US benchmark from C and fails. C is the strategy.

C with a 3- or 12-month lookback also holds: 0.23 and 0.30 full sample. All 70 constructions are in the replication package · 4

4 · The rule we kept: universe, signal, sizing, rebalance

Universe

- G10: AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, SEK vs USD, spot plus the 3-month rate differential
- EM: BRL, MXN, ZAR, CLP, COP, PLN, HUF, CZK, KRW vs USD, the nine floating currencies (Topic 2); managed currencies excluded
- Equity: OECD monthly share-price indices in local currency, US index as the benchmark

Signal

- Six-month log return of the country's index minus the US index, both in local currency, so no FX move enters the signal
- Positive: the market beat the US, long the currency next month. Negative: short it
- No ranking, so when the US beats most markets the book is net long dollars. We call that net position the dollar component and measure it separately

Sizing

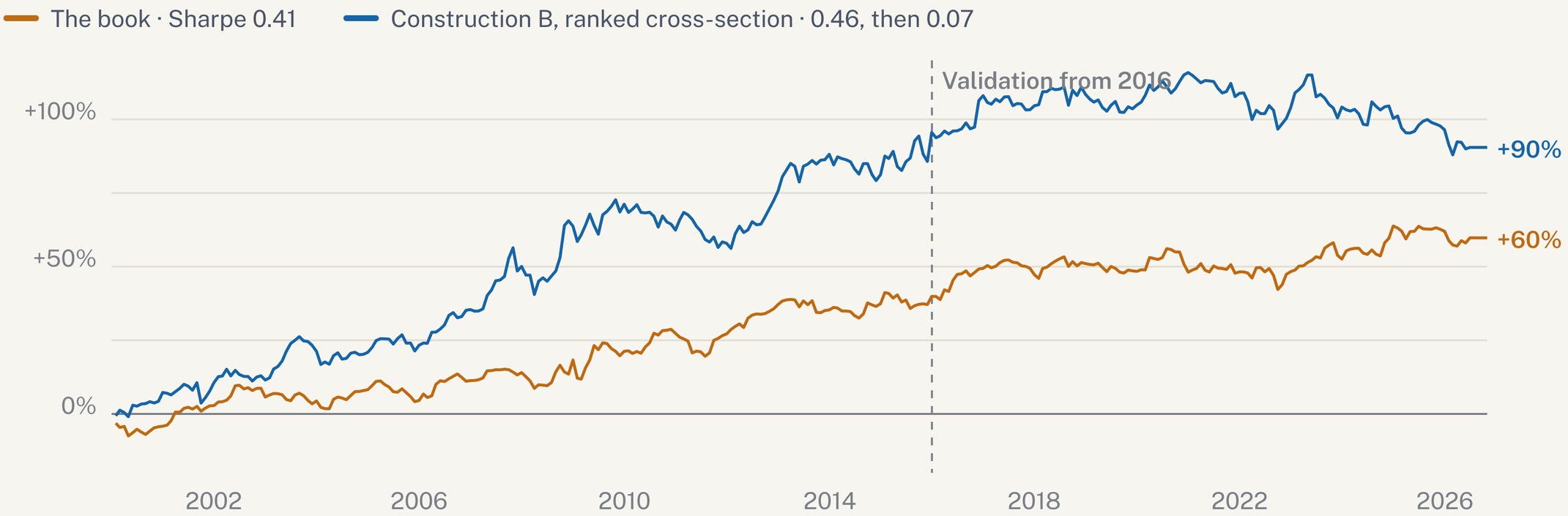
- Equal weight: one-ninth per currency in each book, gross 100%, no leverage, no vol targeting
- G10 runs at about 5% annualised volatility and scales by a constant
- Every currency is always in the book; the signal sets only the sign. Today: G10 long JPY, CAD, NOK, short the other six; EM long BRL, MXN, KRW, PLN, HUF, CLP, short ZAR, CZK, COP

Rebalance and costs

- Month end, for the following month, using indices known at the close; no look-ahead
- Half-spread per unit of turnover: 2 bp in G10, 15 bp in EM. Turnover 3.7× and 3.3× a year, so 7 bp and 50 bp a year
- Two parameters, the 6-month lookback and the US benchmark, both fixed on 2000 to 2015

Spot: FRED H.10 for G10, Brazil, Mexico and South Africa; ECB reference rates crossed through EUR/USD for Poland, Hungary, Czechia and Korea; IMF for Chile and Colombia · 5

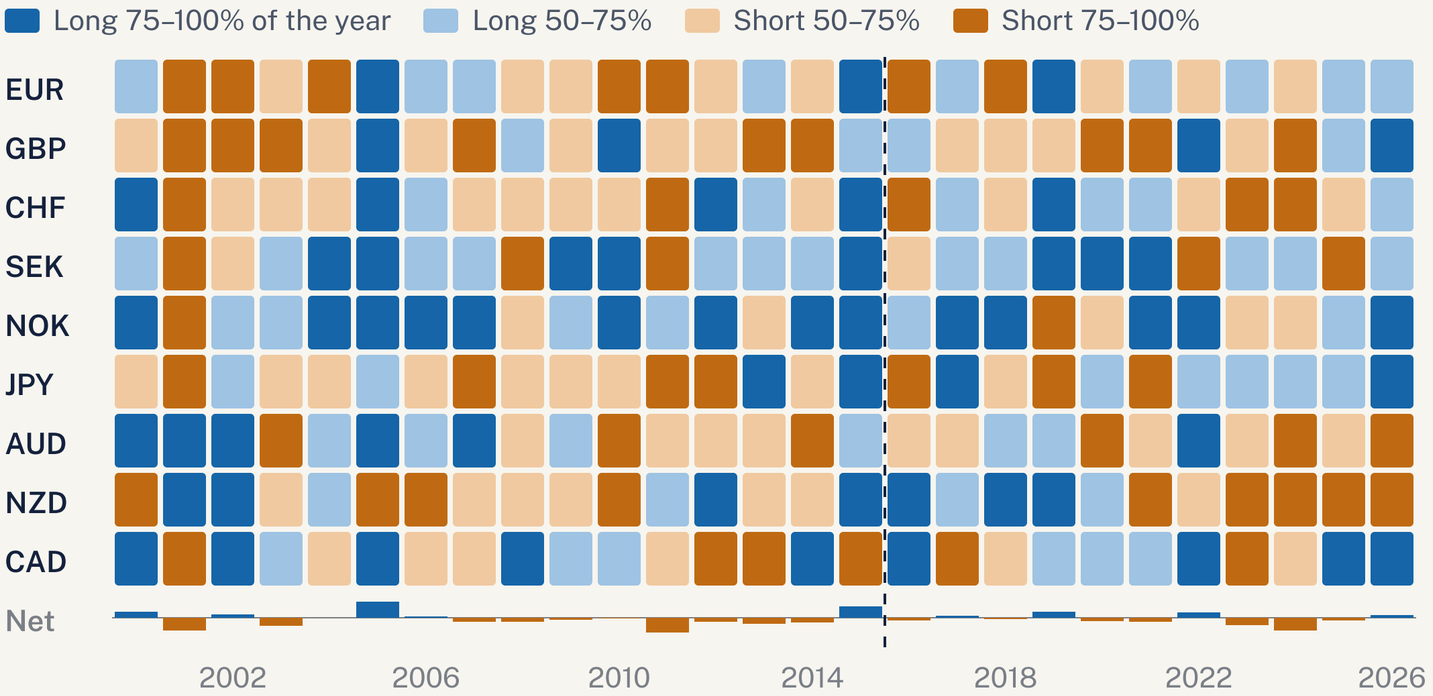
5 · G10 performance after costs, 2000 to 2026



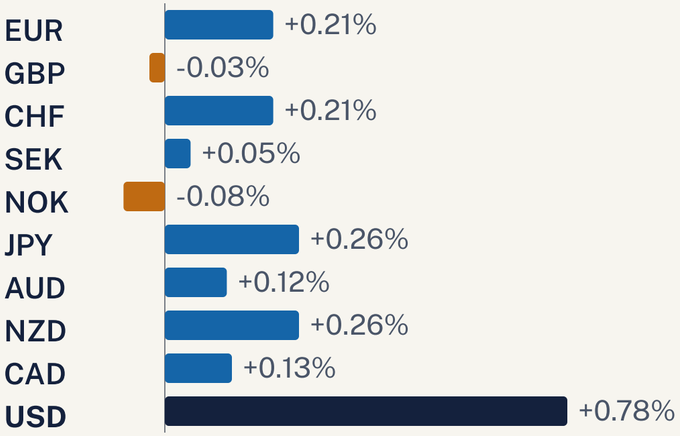
Orange is the book: Sharpe 0.46 before the dashed line, 0.38 after, max drawdown 9%. Blue is construction B, the ranked cross-section: flat since the window it was chosen on ended.

Monthly, nine G10 currencies vs USD, equal weight, excess returns include the 3-month rate differential, 2 bp half-spread per unit of turnover · 6

6 · G10 positions and attribution: every currency flips, the dollar is one leg of ten



ATTRIBUTION BY CURRENCY, % A YEAR

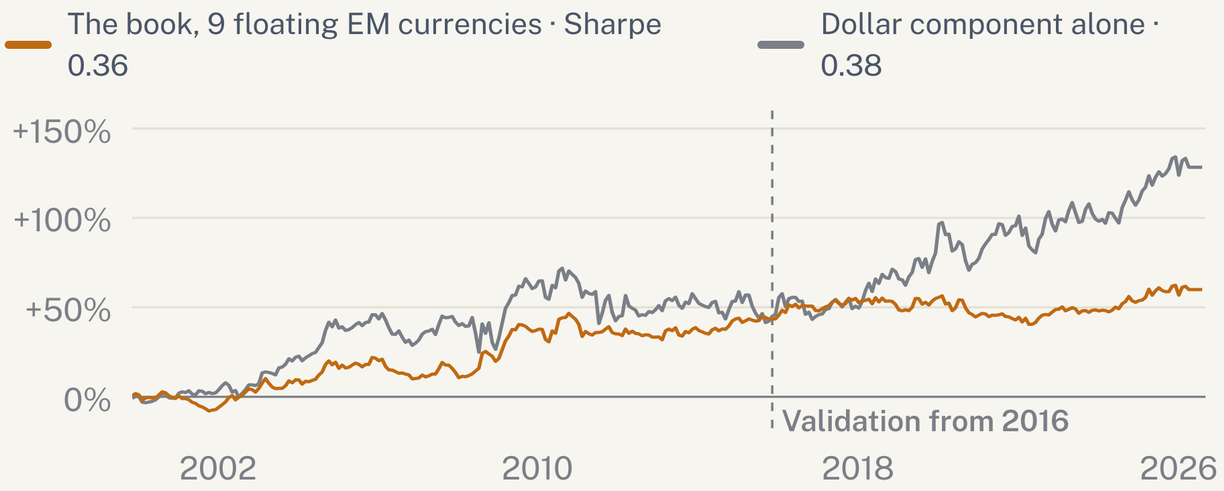


Sum of the ten currencies, gross	+1.92%
Costs	−0.07%
The book, after costs	**+1.85%**

USD is one leg of ten: its exposure is minus the net position, and it earns 0.78% of the 1.92% gross. The other nine earn 1.14% between them, seven of nine positive. No single currency carries the book.

Cells: share of the year's months long vs USD; net row: average net position, −1 = long USD vs all nine. Attribution: each currency's net exposure, USD included, times its return against the equal-weight basket of all ten, gross; the ten sum to the book before costs · 7

7 · Emerging markets, nine floating currencies: mostly the dollar cycle



Four constructions, EM floats, 15 bp	2000–15	2016–26	Full
A · Cross-section, top 3 / bottom 3	−0.27	−0.47	−0.34
B · Cross-section, ranked	−0.06	−0.55	−0.22
C · Dollar-relative vs US: the book	**0.42**	**0.24**	**0.36**
D · Dollar-relative, own return	0.36	−0.04	0.21

Ranking the nine against each other loses money in both halves. Judging each against the US holds in both: 0.42, then 0.24. Dropping the US benchmark fails out of sample, as in G10.

Construction C, nine floats, 15 bp	2000–15	2016–26	Full
The book	**0.42**	**0.24**	**0.36**
Dollar component alone	0.28	0.56	0.38
Dollar component removed	−0.15	−0.22	−0.17
Spot returns only, no carry			0.29
Signal used one month late			0.40

−10% · p = 0.03

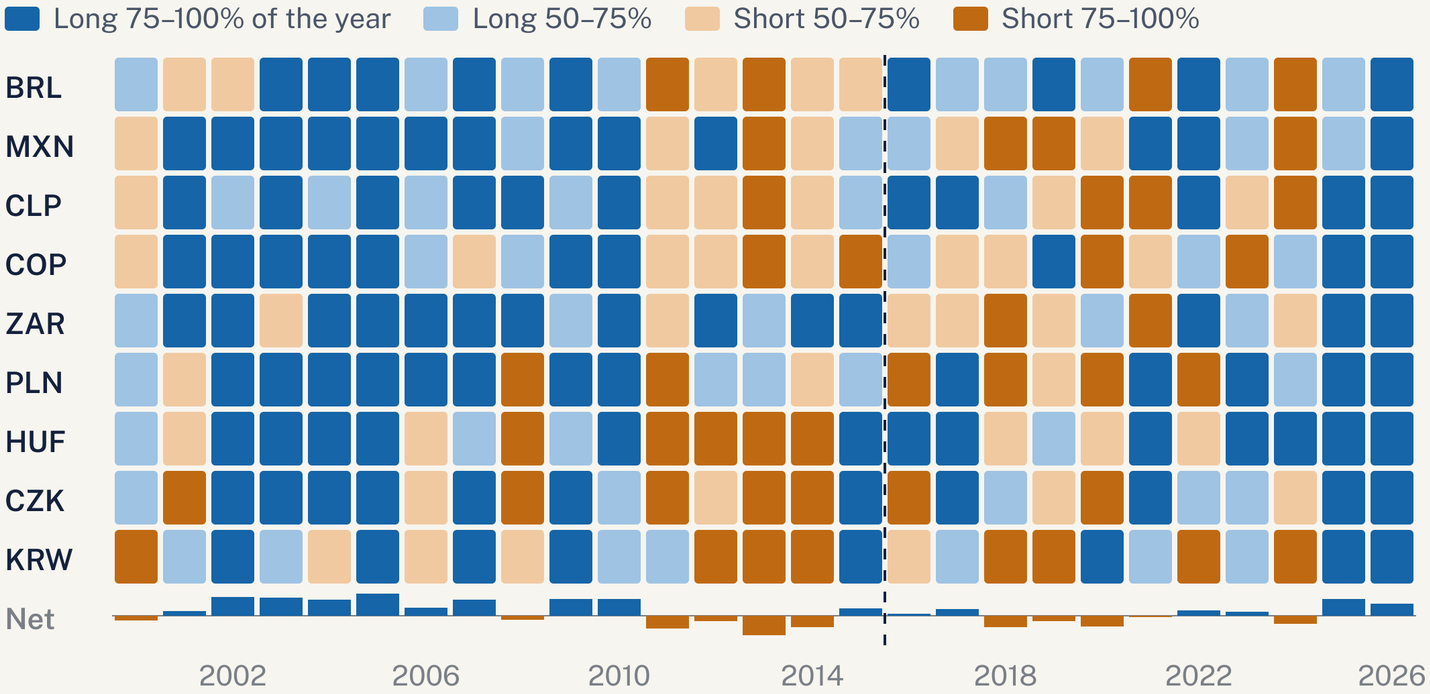
max drawdown, 2000 to 2001; vol 5.3%; 18 of 27 years positive; placebo p against 200 time-shifted books

Mostly the dollar cycle

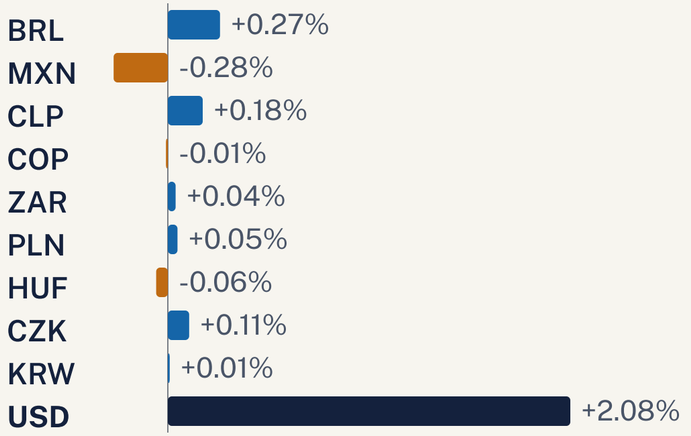
Remove the dollar component and the book loses money. In EM the information is "did the US beat the EM average": risk-on, risk-off. Unlike G10, this is a dollar bet.

Floating per Topic 2: BRL MXN ZAR CLP COP PLN HUF CZK KRW. Managed currencies (INR CNY IDR TRY) excluded: reserves absorb the flow. Rule and lookback fixed on G10. At 2 bp: 0.44 · 8

8 · EM positions and attribution: the dollar leg is nearly the whole return



ATTRIBUTION BY CURRENCY, % A YEAR



Sum of the ten currencies, gross	+2.38%
Costs	−0.49%
The book, after costs	**+1.89%**

USD earns 2.08% of the 2.38% gross: the book is long USD against the basket in 40% of months and short in 60%, and that leg is nearly the whole return. The other nine net 0.30%, four of them negative.

Cells: share of the year's months long vs USD; net row: average net position, −1 = long USD vs all nine. Attribution: each currency's net exposure, USD included, times its return against the equal-weight basket of all ten, gross; the ten sum to the book before costs · 9

9 · Risk: no fat left tail, but long flat stretches, and the dollar is the shock

Distribution, monthly 2000–26	G10	EM floats
Volatility, annualised	4.5%	5.3%
Max drawdown	−9%	−10%
Longest drawdown	36 months	63 months
Worst month	−5.2%	−4.8%
Monthly VaR 95% / CVaR	−1.8% / −2.7%	−2.1% / −2.9%
Skew / excess kurtosis	0.0 / 1.5	0.4 / 1.9
Months positive	54%	52%
Sortino / Calmar	0.63 / 0.21	0.61 / 0.18

Dollar exposure and checks	G10	EM floats
Correlation with broad dollar index	0.03	0.01
Beta to the dollar index	0.03	0.01
Average net foreign-currency exposure	−5% of gross	+13% of gross
All nine on one side, share of months	2.5%	11%
Dollar component alone, Sharpe	0.13	0.38
Book with dollar component removed	0.42	−0.17
Placebo, signal shifted in time, 200 draws	p = 0.03	p = 0.03
Correlation between the two books	0.15	

Shape

Skew is zero or positive and the worst month is about one vol, so this is not carry: no small gains with rare large losses. The cost is time, not depth: three years under water in G10, five in EM, on an unlevered book earning under 2% a year.

When it loses

When the dollar jumps against the net position: Jan 2009 and Sep 2022, net long foreign currency into a dollar spike; Apr 2011, net short all nine as the dollar fell. The signal looks back six months and has no view on the dollar itself.

Dollar from FRED DTWEXBGS. VaR and CVaR are historical, monthly, 95%. Placebo shifts the whole signal matrix forward by a random 12 to 200 months and reruns the rule · 10

WHAT WE WANT YOU TO REMEMBER

Stock-market winners attract flow, and the flow shows up in the currency over the next month.

The version that works is the simplest one: did the market beat the US over six months. In G10 it is country selection, not a dollar bet. In emerging markets, with floating currencies, it is mostly the dollar cycle. The ranked versions looked better in sample and worse out of it.

0.46 → 0.38

G10 Sharpe in the selection window and after it, with a 9% maximum drawdown

0.70 → 0.07

construction B, the ranked cross-section, selection window and after: the lesson about choosing in sample

0.38 vs −0.17

EM floats: the dollar component alone versus the book with it removed. In emerging markets the flow story is the dollar cycle